# Nathnael Dejene

**AI & Computer Vision Engineer**
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## SUMMARY

AI and computer vision engineer with an M.Sc. in Computer Engineering and a strong software engineering foundation. Experienced across the full model lifecycle (training, evaluation, optimization and deployment), with a focus on reliable, well-tested systems that perform under real-world constraints. Comfortable moving between research and production, and explaining technical decisions clearly.

## TECHNICAL SKILLS

**Languages:** Python, C++, SQL, JavaScript, Java, NodeJS, Rust
**ML & computer vision:** PyTorch, TensorFlow, YOLOv8, model evaluation, quantization & pruning, federated learning
**Inference & deployment:** ONNX Runtime, OpenVINO, TensorRT, FastAPI, Flask, REST APIs, Docker
**Engineering practice:** pytest, GitHub Actions CI, mypy, Git, Linux, benchmarking, code review, technical writing

## EXPERIENCE

**Computer Vision Engineer — Ethiopian Artificial Intelligence Institute (EAII)** Jul 2025 – Present

- Build and optimize computer vision systems in Python and PyTorch, and the FastAPI services that deploy them.
- Evaluate model outputs to isolate failure cases and improve reliability before deployment.
- Conduct applied computer vision research alongside engineering work, moving prototypes into production.

**Behavioral Data & Architecture-Strategy Annotation — Revelo (Remote)** 2025

- Rated how AI models approach and solve technical problems against detailed expert rubrics.
- Wrote expert-level reasoning for machine learning tasks used to train and evaluate AI models.

**Engineering Intern — Gibe III Hydroelectric Power Station** Jul 2024 – Sep 2024

- Designed a microcontroller (C++) temperature controller for the generator excitation (SCR) cabinet: an RTD sensor switches cooling fans via relays between high and low set points, even after shutdown. Simulated in Proteus and built to run alongside the existing controller without modifying its proprietary software.
- Worked hands-on with SCADA, PLC and excitation control systems for power-generation equipment.

## PROJECTS

**Vision Inference Microservice** — *FastAPI, TensorRT, OpenVINO* github.com/nati2126/vision-inference-microservice

- Built a YOLOv8 detection API that serves one endpoint from PyTorch, ONNX Runtime, OpenVINO or TensorRT, with a reproducible accuracy, latency and memory benchmark across backends.
- TensorRT FP16 reached 13.8 ms per image (72 img/s), 6.7× faster than PyTorch on CPU with no accuracy loss.
- Traced a naive INT8 model scoring 0 mAP to box coordinates and class scores sharing one quantization scale; keeping the decode tail in FP32 recovered 0.430 mAP50-95 (FP32: 0.444) at 3.5× smaller size.

**BEFLearn: One-Shot Nested Pruning for Federated Learning** — *M.Sc. thesis, PyTorch, Flower* github.com/nati2126/beflearn

- Designed server-side, one-shot nested pruning that sizes each client's model to its bandwidth, cutting total communication by 69.3% and the slowest client's upload from 48.9 to 3.1 min, within 1 pp of unpruned accuracy (CIFAR-10, VGG-11-BN).
- Nested sandwich training raised the two smallest sub-models' standalone accuracy from 15% and 30% to 61% and 62% at no extra communication cost.

**Thinking in Loops: Universal Transformer** — *PyTorch, Gradio* github.com/nati2126/thinking-in-loops

- Built a transformer that solves 4×4 Sudoku by reapplying one shared block and learning when to stop; with 204K parameters it matches an 8-layer, 1.6M-parameter model (99.93% vs 99.95% solved, 3 seeds).
- Showed the halting head is well calibrated; a tuned stopping threshold keeps full accuracy at about a third of the compute.

**MiniTor: Onion Routing Network** — *Python, cryptography, TCP* github.com/nati2126/MiniTor

- Built a Tor-style network with RSA-OAEP key exchange, AES-GCM authenticated layers, fixed-size cells, replay protection and exit policies; end-to-end tests run in CI on Linux, Windows and macOS.

## EDUCATION

**M.Sc. in Computer Engineering — Addis Ababa Science and Technology University** 2024 – Aug 2026
*Thesis: One-Shot Nested Pruning for Resource-Constrained Federated Learning*

**B.Sc. in Electrical and Computer Engineering — Addis Ababa Science and Technology University** 2020 – Jun 2025
Relevant coursework: Data Structures & Algorithms, Computer Vision, Networks, Distributed Systems, Operating Systems.